Sec. 2.3 & 2.5

Modeling with D.E.'s

① Sum of money is deposited in a bank, with rate r

$S(0) = 0$ initial investment

Interest compounds once per year

S_0

1st $S_0 + rS_0 = S_0(1+r)$

2nd $S_0(1+r) + rS_0(1+r) = S_0(1+r)^2$

⋮

$S(t) = S_0(1+r)^t$

If: the interest is compounded m times per year after t years

$\rightarrow S(t) = S_0\left(1+\frac{r}{m}\right)^{mt}$

If: the interest is compounded continuously

then $m \to \infty$

$\rightarrow S(t) = S_0 \lim_{m\to\infty}\left(1+\frac{r}{m}\right)^{mt}$

$= - - - \left(1+\frac{1}{\frac{m}{r}}\right)^{\frac{m}{r}rt}$

$\left[\text{let } n = \frac{m}{r}\right] \rightarrow = S_0\left[\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n}\right)^{rt}]$

$= S_0 e^{rt}$

$\left(\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e\right)$

$S(t) = S_0 e^{rt}$ is the sol. of $\begin{cases} \frac{dS}{dt} = rS \\ S(0) = S_0 \end{cases}$

Δt is small

$$S(t+\Delta t) = S(t) + S(t) r \Delta t$$

$$S(t+\Delta t) - S(t) = r S(t)$$

$$\left(\Delta t \to \lim_{\Delta t \to 0} \;\middle|\; S'(t) = rS \right)$$

Assume deposits at a constant rate k $\left(= \frac{\text{money}}{\text{time}}\right)$

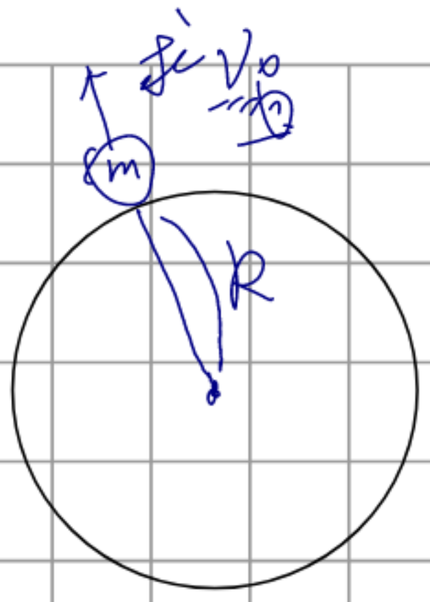
$$\frac{dS}{\Delta t} = \text{rate in} - \text{rate out}$$

$$= rS + k$$

$$\Rightarrow S(t) = S_0 e^{rt} + \frac{k}{r}(e^{rt} - 1)$$

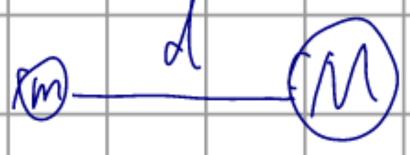
$$S(0) = S_0$$

(2)



$$F = ma = m\frac{dv}{dt}$$ → velocity

(1)



$$F = \frac{kmM}{l^2}$$

$$m\frac{dv}{dt} = \frac{-C}{(R+x)^2} = F \text{, where } C = mgR^2$$

↳ displacement

$$v(0) = v_0$$

$$F_{x=0} = -mg = -\frac{C}{R^2}$$

$$\Rightarrow m\frac{dv}{dt} = \frac{-mgR^2}{(R+x)^2}$$

$$\frac{dx}{dt} = v(t)$$

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = a$$

$$v(0) = v_0$$

$$\frac{dv(x(t))}{dt} = \frac{dv}{dx}\frac{dx}{dt} = \frac{dv}{dx}v$$

$$= \frac{gR^2}{(R+x)^2}$$

$$v_0 = v(x=0) = v(0)$$

$$\left(\frac{v^2}{2} = \frac{gR^2}{R+x} + \frac{v_0^2}{2} - gR\right)$$

$$\int v\,dv = \int \frac{-gR^2}{(R+x)^2}\,dx$$

$$\frac{v^2}{2} = \frac{gR^2}{R+x} + C$$

$x = 0 \qquad \frac{v_0^2}{2} = \frac{gR^2}{R+0} = C$

$$\Rightarrow C = \frac{v_0^2}{2} - gR$$

$$v(x) = \sqrt{v_0^2 - 2gR + \frac{2gR^2}{R+x}}$$

When $v = 0$, x is x_{max}

$$v_0^2 - 2gR + \frac{2gR^2}{R+x} = 0$$

$$\Rightarrow x_{max} = \frac{v_0^2 R}{2gR - v_0^2}$$

Escape velocity: when $x_{max} \to \infty$

When $2gR^2 = v_0^2 \Rightarrow v_0 = \sqrt{2gR}$

③ $\frac{dy}{dt} = ry$ (← population ~ time)

$y(0) = y_0$

$y(t) = y_0 e^{rt}$

$\frac{dy}{dt} = y(r - \alpha y)$

2 Population Growth Models Sec. 2.5

$$\frac{dy}{dt} = ry$$

$$y(0) = y_0$$

$y(t) \geq 0$

$y(t) = y_0 e^{rt}$

$$\begin{cases} \frac{dy}{dt} = y f(r,y) \\ y(0) = y_0 \end{cases}$$

$f(r,y) = r - ay$

② $\frac{dy}{dt} = y(r-ay)$

$= ry\left(1 - \frac{a}{r}y\right)$

$= ry\left(1 - \frac{y}{k}\right)$, where $k = \frac{r}{a}$

$\frac{dy}{dt} = f(y)$, $f(y)$的符号取决于 the monocity of $y(t)$

$f(y) = ry - \frac{r}{k}y^2$

$f'(y) = r - \frac{2r}{k}y$

$f''(y) = -\frac{2r}{k} < 0$

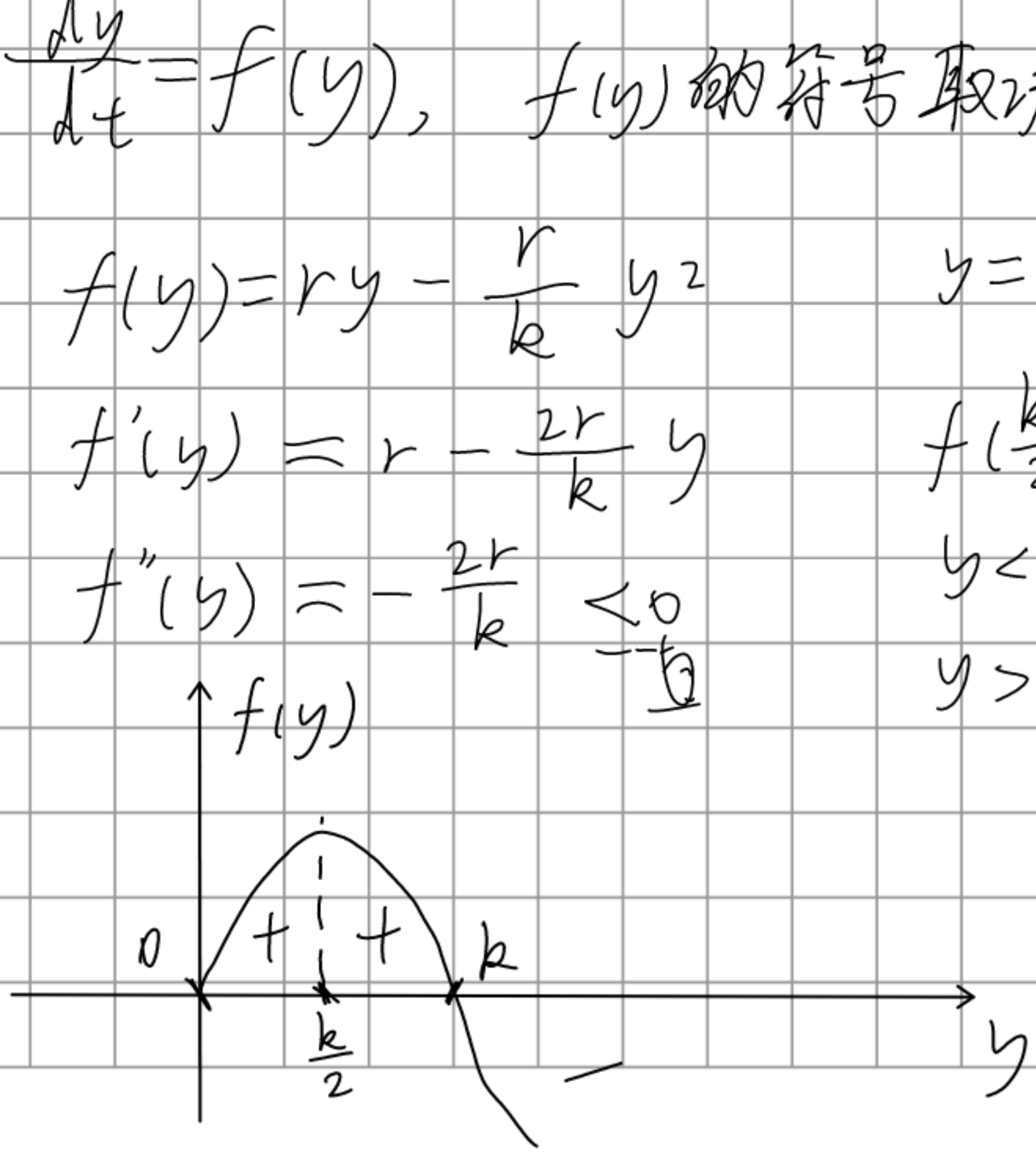
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$y = \frac{k}{2} \Leftrightarrow f'\left(\frac{k}{2}\right) = 0$

$f\left(\frac{k}{2}\right) = \cdots = \frac{rk}{4} > 0$

$y < \frac{k}{2}$, $f' > 0$, $f \uparrow$

$y > \frac{k}{2}$, $f' < 0$, $f \downarrow$



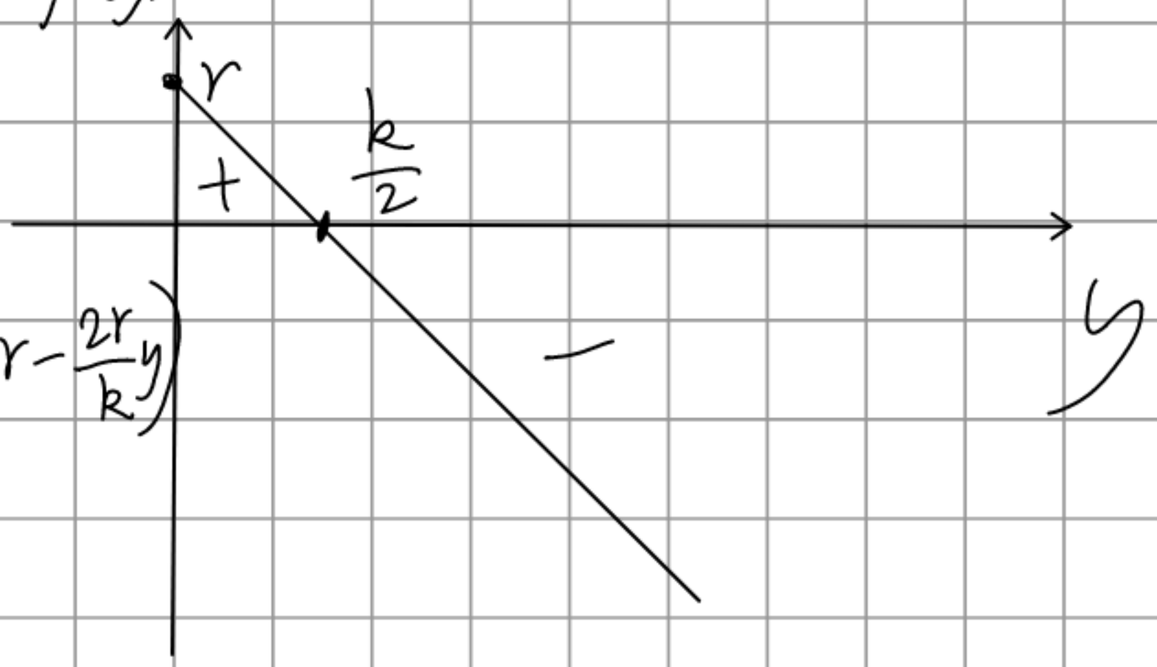
$y=0$ & $y=k$ are called equilibrium solutions

$$\frac{d^2y}{dt^2} = \frac{d}{dt}\left(\frac{dy}{dt}\right)$$

$$= \frac{d}{dt} f(y) = f'(y)\frac{dy}{dt}$$

$$= f'(y) f(y)$$

$f'(y)$



$\left(f'(y) = r - \frac{2r}{k}y\right)$

接下來: Solve the eq. $\frac{dy}{dt} = ry\left(1 - \frac{y}{k}\right)$

$y = y_1 + \frac{1}{u} = k + \frac{1}{u}$, $\frac{dy}{dt} = -\frac{1}{u^2}\frac{du}{dt}$

$-\frac{1}{u^2}\frac{du}{dt} = r\left(k + \frac{1}{u}\right)\left(1 - 1 - \frac{1}{ku}\right)$

$= -r\left(k + \frac{1}{u}\right)$

$\frac{1}{u^2}\frac{du}{dt} = \frac{r}{k\cancel{u}}\left(k + \frac{1}{u}\right)u^{\cancel{2}}$

$= \frac{ru}{k}\left(k + \frac{1}{u}\right)$

$= ru + \frac{r}{k}$

$e^{-rt}\left(\frac{du}{dt} - ru\right) = \frac{r}{k}e^{-rt}$

$\frac{d}{dt}\left(e^{-rt}u\right) = \frac{r}{k}e^{-rt}$

$e^{-rt}u = \int \frac{r}{k}e^{-rt}\,dt$

$= -\frac{1}{k}e^{-rt} + C$

$u(t) = Ce^{rt} - \frac{1}{k} = \frac{kce^{rt} - 1}{k}$

$\Rightarrow \frac{1}{u} = \frac{k}{kce^{rt} - 1}$

$y(t) = k + \frac{1}{u}$

$= k + \cdots$

$\Rightarrow y_0 = y(0) = k + \frac{k}{kc - 1}$

$\Rightarrow c = \frac{1}{k} + \frac{1}{y_0 - k}$

$$\left(\Rightarrow y(t)=\frac{y_0 k}{y_0+(k-y_0)e^{-rt}}\right)$$

(2) $\dfrac{dy}{dt}=f(y)$

$$\int \frac{dy}{f(y)} = t+C$$

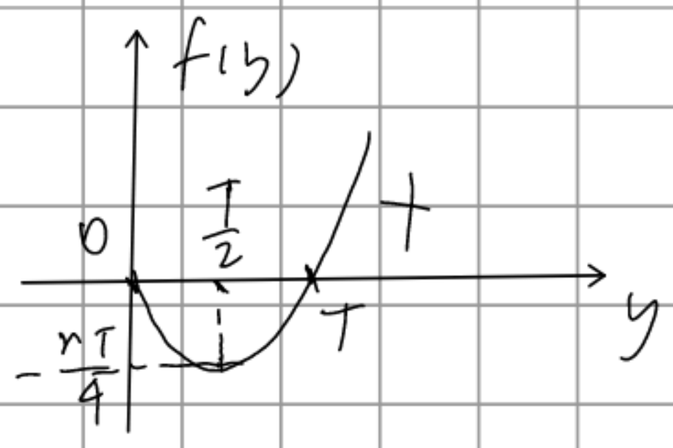
(3) $$\begin{cases}\dfrac{dy}{dt}=-ry\left(1-\dfrac{y}{T}\right)\\ y(0)=y_0\end{cases}$$

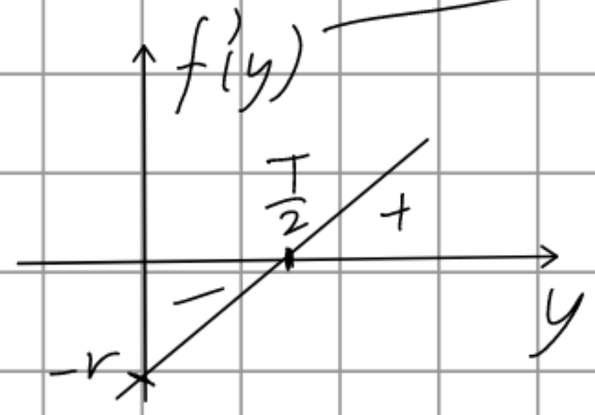
$$f(y)=-ry+\frac{ry^2}{T}$$

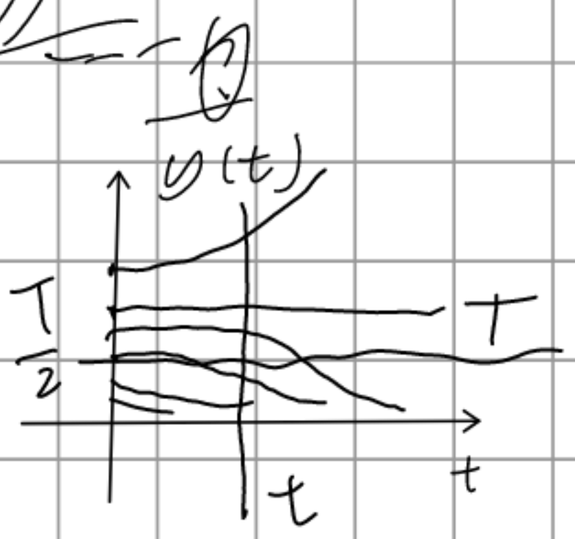
(2)

$$\Rightarrow f'(y)=-r+\frac{2ry}{T}$$

$$\frac{dy}{dt}=f(y)\Rightarrow \frac{d^2y}{dt^2}=f(y)f'(y)$$







$$f\left(\frac{T}{2}\right)=\cdots=-\frac{rT}{4}$$

Sol: $y(t)=\dfrac{y_0 T}{y_0+(T-y_0)e^{rt}}$

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- — — (★)

$$\frac{dy}{dt} = -r\left(1 - \frac{y}{T}\right)$$

$$y(0) = y_0$$

$$\lim_{t \to t^*} y(t) = \infty$$

$$t^* = \frac{1}{r} \ln \frac{y_0}{y_0 - T}, \quad y_0 > T$$

$$y(t) = \frac{y_0 T}{y_0 + (T - y_0) e^{rt}}$$

$$y_0 + (T - y_0) e^{rt^*} = 0$$

Logistic Growth with a threshold

$$\frac{dy}{dt} = -ry\left(1-\frac{y}{T}\right)\left(1-\frac{y}{k}\right), r>0,$$

$$0<T<k$$

$$y(0)=y_0$$

$$f(y)=-ry\left(1-\frac{y}{T}\right)\left(1-\frac{y}{k}\right)$$

$$f'(y)=-r+2yr\left(\frac{1}{T}+\frac{1}{k}\right)-\frac{3y^2r}{Tk}$$

Inflection points

$$f'(y)=0 \Leftrightarrow 3y^2-2(k+T)y+Tk=0$$

$$\frac{1}{4}k^2+T^2-Tk=\left(k-\frac{T}{2}\right)^2+\frac{3}{4}T^2>0$$

$$\Delta=4(k+T)^2-12Tk=4(k^2+T^2-kT)$$

$$y_{1,2}=\frac{2(k+T)\pm 2\sqrt{k^2+T^2-kT}}{2\cdot 3}$$

$$y_+>0,\ y_->0,\ 0<y_-<y_+<k$$

$$T<\frac{k+T+\sqrt{k^2+T^2-kT}}{3}<k,$$

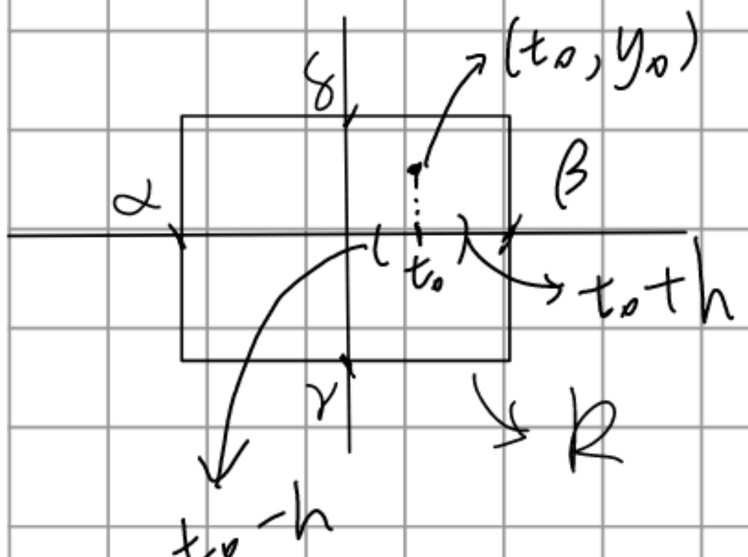
$$(2k-T)^2>k^2+T^2-kT$$

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{k}\right)$$

$$y = k + \frac{1}{u} \quad \text{---} ①$$

Sec. 2.4 and 2.8

Thm.: Let $f(y,t)$ and $\frac{\partial f}{\partial y}(y,t) = f_y(y,t)$ be cont. in some rectangle:



$$\begin{cases} \alpha < t < \beta, \quad \gamma < y < \delta, \\ \text{含点 } (y_0, t_0) \end{cases}$$

Then $\exists$ interval (t_0-h, t_0+h) and t_0,

$h > 0$,

s.t. $\exists$ unique sol. t_0,

$$\begin{cases} \frac{dy}{dt} = f(y,t) \\ y(t_0) = y_0 \end{cases}$$

↘ 定理!

$$\begin{cases} y' + p(t)y = g(t) \\ y(t_0) = y_0 \end{cases}$$

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t) g(t)\, dt + C\right],$$

where $\mu(t) = e^{\int p(t) dt}$

$f(y,t) = g(t) - p(t)y$